The friction surface lab

Teacher guide and worked answers

Compare a sliding block’s movement on different surfaces and explain the effect of friction using a controlled test.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Summer | C36

Compare how things move on different surfaces; explore friction (push and pull).

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

A push or pull can change movement.

A comparison needs a consistent measurement rule.

Success evidence

Describe friction opposing this block’s sliding.

Compare repeated run-out distances.

Name the changed surface and important controls.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Optional low stable ramp with a smooth flush transition to a tabletop tray; small wooden sliding block; card, cloth and rubber test strips; ruler; barrier to catch the block.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Adult pilots a low ramp so the block slides without tumbling or snagging and stops within the tray. Fix the ramp and transition; use the same block/contact face and marked release point without a push.

The supplied constructed distances allow the full lesson without equipment. For real trials measure from the ramp end to the same trailing block edge at stopping; use three releases per surface and keep actual results separate.

Practical care

Keep the activity on a stable table in a contained tray with a stop barrier. No pupils sliding on floors or ramps.

Use a small block, low ramp and clear release area; keep hands out of its path. No heavy falling objects or stretched elastic launchers.

Do not lubricate the surfaces or create a floor-slip hazard. Stop any setup that tips, snags or lets the block leave the tray.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Which surface should stop the sliding block sooner?
2	Retrieval	2-5	Retrieve pushes, pulls and comparisons
3	I do	5-12	I distinguish motion from force
4	I do	Within stage	I keep the start conditions fixed
5	I do	Within stage	I use repeats before choosing
6	We do	12-16	The surface-selection meeting
7	We do	16-20	Repair a changed release
8	Hinge	20-23	The block slides right after release.
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Which surface should stop the sliding block sooner?

Set purpose: choose a surface for a small model stopping zone, not for real vehicle safety equipment.

2 Retrieval Retrieve pushes, pulls and comparisons

1 motion/speed/direction. 2 relevant control accepted. 3 cm. Explain that a block already moving can continue after a hand stops pushing.

3 I do I distinguish motion from force

Think aloud: “The direction of movement is not automatically the direction of every force. Here friction opposes the sliding and slows the unpushed block.” This is not a complete force diagram: weight and support act vertically.

4 I do I keep the start conditions fixed

Shares I do time. Think aloud: “A higher release might travel further even on the same material. I keep the ramp and release fixed before comparing surfaces.” Fix surface strips flat with comparable transitions and no bunching.

5 I do I use repeats before choosing

Shares I do time. Think aloud: “Card is around 60 cm, rubber around 15 cm. In this setup rubber gives shorter run-out. I do not claim the material name gives the same result with every object.” Means are supplied summaries; core pupils can compare repeated ranges.

6 We do The surface-selection meeting

W1: rubber gives shortest run-out (14-16 cm). Partners take client and evidence adviser roles; client challenges a claim with a table reading.

7 We do Repair a changed release

W2: no; a push changes initial conditions. Repeat using the same marked release without a push; retain/annotate the flawed run rather than silently replacing it.

Reveal: Use comparable starts and record any method change.

8 Hinge The block slides right after release.

Correct first option. If second chosen, point to relative sliding at the contact.

A. Friction on the block acts left, opposing sliding. - It opposes relative sliding on the stationary surface.

B. Friction acts right because the block moves right. - Movement and friction point in opposite directions here.

C. No force can affect the block after the hand lets go. - Contact friction can continue after release.

9 You do Choose your science route

Set ONE route. Use supplied readings for the core task; optional practical after adult setup check. Keep actual values separately labelled. Do not require forces equations, coefficients or mean calculation for BUILD.

10 You do Make your own evidence record

Use supplied readings for the core task; optional practical after adult setup check. Keep actual values separately labelled. Do not require forces equations, coefficients or mean calculation for BUILD.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Use supplied readings for the core task; optional practical after adult setup check. Keep actual values separately labelled. Do not require forces equations, coefficients or mean calculation for BUILD.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: Comparable starts, repeated distances and a purpose-specific claim make the choice stronger.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: Comparable starts, repeated distances and a purpose-specific claim make the choice stronger.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 movement/speed/direction. 2 relevant controlled condition. 3 cm.

We do W1-W2

W1 rubber, 14-16 cm compared with card 58-62 cm and cloth 22-24 cm; it fits shorter run-out here. W2 an extra push changes starting conditions; repeat from the same release mark without a push and annotate the flawed run.

Hinge

First option correct. Friction on the right-sliding block points left; contact force persists after release.

S1-S5

S1 rubber. S2 14,15 or16 cm. S3 left. S4 same block/contact face, ramp/release point, transition, distance rule or other relevant condition. S5 no: changes start conditions.

M1-M4

M1 change horizontal surface; measure run-out distance in cm. M2 card 58-62 cm; rubber 14-16 cm, rubber shorter in every run. M3 friction opposes relative sliding, acts left on right-moving block and slows it after release. M4 fix ramp and mark; same block/face; release without push; measure from ramp end to same trailing edge at rest; repeat three runs per surface and retain records.

T1-T4

T1 rubber gave the shortest run-out for this wooden block/contact face and controlled setup. T2 friction depends on the pair, surface condition and loading; a different use needs relevant evidence. T3 card, about 60 cm here. T4 record the bunching/snag and affected run, secure the strip flat with comparable transition, repeat under the stated method and distinguish snagging from normal sliding resistance.

Exit E1-E3

E1 relative sliding between block and surface. E2 rubber. E3 fixed ramp angle/release mark, same block face, no push or comparable transition.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Horizontal run-out is a comparison of slowing in this apparatus, not a direct coefficient-of-friction measurement. Ramp losses and initial conditions must remain comparable.

Secure core: correct friction direction for this setup, table-supported comparison and one meaningful control. The lesson does not recommend surfaces for real safety-critical equipment.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

Friction is always useless.

Friction can provide useful grip or slow sliding; usefulness depends on the purpose.

Check: Ask whether a surface should stop or allow the block to slide.

The surface with the longest run-out has the most slowing effect.

Under the same release conditions, greater resistance usually reduces run-out.

Check: Compare card and rubber readings.

A smooth-looking material always has least friction.

Friction depends on the pair of materials and conditions; appearance alone is insufficient.

Check: Ask for test evidence rather than a universal texture rule.

Word	Meaning
friction	A contact force that opposes sliding between surfaces.
surface	The outside layer in contact with an object.
run-out	Distance travelled along the level test surface after leaving the ramp.
fair comparison	A comparison with relevant conditions kept the same.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

OpenStax: friction

Checked 8 September 2026. Sliding friction opposes relative sliding at the contact. In a horizontal constant-speed pull, opposing horizontal forces balance. Surface-pair results are not universal rankings.